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1. INTRODUCTION

Durban University of Technology (DUT) is the first choice for higher education in Kwazulu-Natal with 33 000 students. The DUT community has grown for years, making the safety of students/staff and university assets a big priority for the institution. Given the importance of safety the institution has a sub-unit under Protection Service Unit (PSU) namely Card Production. The main responsibility of card production is to produce access cards for students, staff, and relevant sub-contractors to enable easy access control to the institution. Due to the long queue problem faced by the institution during the card production period we decided to conduct an investigation to study the processes of the business to understand the roots of the problem of long queues so that we could help the department with recommendations/solutions to the problem at hand. The presentation of this business case is critical to show the need for improvement in the card production current strategy, the case outlines the current system challenges/limitations along with proposed solution or recommendations to address the problem at hand. Additionally, it includes the costs and benefits of implementing the solution, risks, and counter measures to mitigate risks and finally the decision that the management is required to take.

2. EXECUTIVE SUMMARY (Wright, 2008)

- **What the study was about and the findings**

The study aimed to evaluate the viability of the business processes within the card production department, through the analysis of its business operations, technical feasibility, and financial feasibility in meeting the university staff and student's demands. The current software, Card5, used by the department depends on outdated technology, resulting in slow processing times, limited automation capabilities, and compatibility difficulties with newer software and hardware systems. Additionally, insufficient space within the card production department worsens the situation, leading to overcrowding and prolonged wait times for students and staff who want to make their access cards.

We looked into a number of solutions, such as an online acquisition system, updating the physical infrastructure and implementing a card refresh system, to address these

issues. With the card refresh solution, access card management is simplified and students only need to create their cards once, with automatic reactivation in subsequent years. Even though this approach has benefits like faster processing times and maximum effectiveness during busy times, it comes with a very large initial setup and maintenance costs for the database.

Upgrading the physical infrastructure of the card production department allows increased productivity and user experience. This solution includes renovating facilities and installing advanced equipment that will enhance operations at the card production department. Even though the upgrade might lead to long-term cost reductions and increased efficiency, it also poses challenges such as potential disruptions during installation and integration with existing systems and not to forget the huge capital needed for the upgrading process.

Nevertheless, after careful examination, we concluded that the online card acquisition system offers the most practical way to deal with the issues raised. Among the many advantages of this system are its streamlined procedures, shorter wait times, and improved accessibility for both staff and students. Furthermore, it is consistent with our company's goals of using technology to cut expenses and increase productivity. To successfully address the current issues, we advise giving the online card acquisition system's implementation top priority.

- **Advantages:**

Enabling staff and students to remotely acquire the access card through the online system simplifies the card acquisition process. As a result, less manual data entry is required, saving the department and the people involved time and resources. The online system is convenient because it is accessible to staff and students from any location with an internet connection, offering increased flexibility and convenience. There are fewer obstacles to getting their access cards because they can finish the registration process self-paced without having to go to the physical location. A lot of requests can be processed at once by the system, which ensures quick access card issuance and processing, if for any reason the system can't handle the request, it can be upgraded. By moving the access card acquisition online, the card production department experiences lesser queues and decreased congestion, which improves the overall experience for all parties involved.

- **Disadvantages:**

Not all staff or students have access to the internet, or the knowledge and skills required to use the online system efficiently. People who are less tech-savvy or have restricted access to technology or the internet will encounter difficulties when this system is introduced.

Data security and privacy are issues that are brought up when storing private documents and images online. If sufficient security measures are not put in place and kept up to date, there is a huge chance of data breaches or unauthorized access to confidential information from scammers who would use the information and pictures for malicious purposes.

- **The recommendation being made, and decision required.**

Our study and findings lead us to suggest that the Card Production Department at Durban University of Technology should implement an online automated card acquisition system that will allow the students and staff to register and create their access cards online anywhere where there is internet connection. Aside from increasing efficiency, this system will improve the overall experience for staff and students by streamlining the card acquisition process. The Card Production department in collaboration with the university needs to take immediate action and act on the proposed solution because that will benefit not only them but the students and staff.

3. The Business Need

3.1 SWOT ANALYSIS

SWOT analysis is a strategic planning tool that is used to identify and analyse the internal strengths and weaknesses, as well as external opportunities and threats, that affect an organization's performance. It divides these variables into four categories, offering an organized framework for decision-making. Organizations can understand their place in the market better, evaluate opportunities and risks, and create plans to strengthen their positions and address their weaknesses by using SWOT analysis.

For our assignment, we opted for SWOT analysis because it provides a methodical way to assess the organization's external and internal influences, enabling well-informed decision-making. In conclusion, SWOT analysis improves organizational performance and competitiveness by helping set goals, plan strategically, and pinpoint areas needing improvement. Its simplicity and efficiency make it widely used across multiple industries for guiding strategic decision-making (GURL, 2017).

Strengths:

- Skilled employees.
- Openness to investigating innovative solutions.
- Long-lasting and dependable hardware that ultimately saves money.
- Dedication to resolving concerns with service quality.

Weaknesses:

- Outdated technology is negatively affecting operational efficiency.
- Low staffing numbers affect the provision of services.
- Lack of modernization initiatives.
- Employee burnout caused by excessive workload.
- Inability to meet peak demand periods effectively.
- Limited ability to manage increasing numbers of registrations.

Opportunities:

- Introduction of new technology.
- Expansion of service offerings to meet diverse needs.
- Working together with technology partners for innovation.
- Development of an online system to streamline the processes.
- Invest in staff training and development.

Threats:

- Competition from other institutions.
- Danger of data breaches brought on by outdated security protocols.
- Bad publicity brought on by inefficiencies or service delays.
- Possible departure of employees or students as a result of dissatisfaction.
- unanticipated events causing disruption.

3.2 VALUE CHAIN ANALYSIS

Value chain analysis involves mapping and analyzing the full range of activities needed to bring a product or service from conception to its final disposal. It aids in identifying important players, their functions within the value chain, and the relationships and rival channels they face. Value chain analysis is used to understand market dynamics, optimize processes, and identify opportunities for improvement. The reason we selected value chain analysis was its capacity to offer valuable perspectives on market dynamics, stakeholder connections, and possible areas of intervention, thereby supporting well-informed decision-making and strategic planning. Its integration of quantitative and qualitative tools enables a thorough comprehension of the dynamics of intricate market systems (Meijer, 2006).

Primary Activities

- **Inbound Logistic**

Sourcing of materials for producing access cards such as printing supplies, Ink, PVC sheets, RFID chips, Overlay films etc. A good communication and managing relationship with suppliers lead to timely delivery of materials. Additionally, they leverage economies of scale to utilize bulk procurement in order to obtain competitive prices for raw materials.

- **Operations**

The card production unit is currently based solely in Ritson Campus, serving as the sole location for supplying cards to students and staff across four different Campuses. The access cards are designed using the ITS Card5 software and then further printed with Fargo Hdp5000 printer, if the student is a first year, they must take a picture using a webcam. The layout of the access card contains name of the institute, student information captured from the ITS, A photo for identification and security features.

- **Outbound Logistics**

Employees in the Department create and issue cards to students and staff immediately upon request.

- **Marketing and Sales**

There is minimal to none effort requirement for marketing in this field of work. The staff and students receive communication through their emails when it is time to renew their access cards. Support/Assistance is provided to students and staff who come across challenges with their cards for example card loss or technical problems.

- **Service**

The primary objective of the card production unit is to produce both student and staff cards to facilitate easy access control at the DUT facilities and Campuses. Support is provided through telephone and e-mail queries.

Secondary Activities:

- **Infrastructure**

The Other departments/stakeholders working with the card production department are finance, student housing, student admission, different faculty offices, ITS, and stores.

- **Technological Development**

The current system is easy to navigate hence they hire DUT alumni to help during peak times. The university system is currently migrating from ITS to SAP.

- **Procurement**

The Card production unit purchases all its raw material needed for production from Stores, materials include, ink and printing supplies, laminating material, PVS sheets etc.

3.3 Analysis of the business structure and culture

Business Structure: Being one of the 5 sub-units under the protection services department at Durban University of Technology, the card production structure is a hierarchy consisting of a production manager who is responsible for the overall business processes, an ITS support team that handles the technological systems within card production like Card5 software and Databases relevant for this type of productivity. production supervisor who oversees the day-to-day operations, and lastly the administrative staff who operates the Card5 system and equipment such as webcams; printers; etc.

business culture: The culture within the department is likely almost the same as that of the whole Institution which is guided by the principles of fairness, professionalism commitment, compassion, and excellence. for effective productivity the card production unit must also adhere to quality standards by ensuring that effective security features, readability, and durability. Accuracy and consistency standards to ensure reliability by accurately encoding data and personalizing. There are also incentives for staff within the department if they reach targets and for effective team effort, this culture ensures effective work environments and increased productivity.

Levels of Formality: The Card Production units deals with a crucial part of the university which involves effective security and management of individuals who access the facilities thus it is important that they handle the work with utmost care. one of the principles guiding the operation is Professionalism which requires them to demonstrate competency, diligent work ethic and effective team effort. Formal record keeping is essential within the Card production for performance evaluation purposes and financial purposes.

Channels of communication: The Unit communicates by means of a telephone line and emails with the relevant departments involved in the production process (Finance, Faculties, ICTs, Student Housing, etc.) and they maintain constant communication with Stores which is the supplier of materials used in the access production process.

Values and Norms: The DUT Envision2030 living values framework guides the Card production unit, the framework is built on some values and principles that play a crucial role in building a thriving and sustainable DUT community these values include Transparency, honesty, accountability, respect, and Integrity. The card production department associates mostly with the value of integrity as they must consistently prioritize accuracy and quality in their work to maintain the integrity of access cards which in turn improves security. Additionally, an emphasis on teamwork and collaboration within the department and with various stakeholders is essential for effective operation in the production process.

3.4 GAP

Current Situation: The manual access card issuance procedures used at DUT result in lengthy lines, staff burnout, and students missing lectures because of the time they spend standing in line. Even though this system works well, the university's efficiency and online presence are noticeably lacking. The difficulties include staff and student dissatisfaction, resource waste, and operational inefficiencies.

Gap Identification: The organization's 2030 vision places a strong emphasis on innovation, creativity, and cutting-edge infrastructure to boost output and involvement in local and global development. These goals are not, however, supported by the current manual access card issuance system. The organization's current manual processes, which hinder efficiency, innovation, and adaptability, clearly fall short of the organization's desired target state as stated in its vision.

Solution: To bridge this gap, DUT needs to implement an online access card acquisition system for the card production department. This solution aligns with the organization's vision by enhancing efficiency, innovation, and adaptability. Through the implementation of online access card registration, which will enable staff and students to pick up their cards from the department, the department can minimize employee burnout, simplify procedures, and enhance overall satisfaction. This solution gives all stakeholders a convenient and user-friendly experience, which reflects the

organization's values of transparency, integrity, and respect. It also supports the values of justice, dedication, and excellence by making investments in cutting-edge systems and technology to improve service delivery and operational efficacy.

4. Describe the current situation

4.1 Examine the capabilities and processes of the organisation.

1. Access Card Department Capabilities

1.1. Card Issuance - The Access Card Department is able to issue physical access cards to staff members and students in accordance with their authorization and requirements.

1.2. Card Activation - This process ensures that newly issued access cards are activated and linked to the user's account, giving them access to authorized areas and services within the organization.

1.3. Card Deactivation - When an individual's access rights are changed or terminated, or the card is lost, stolen, or otherwise compromised, the department has the authority to deactivate the card by suspending or cancelling card privileges.

1.4. Card Replacement - The ability of the Access Card Department to provide new cards with updated credentials in order to replace lost, damaged, or malfunctioning cards and guarantee users' continuous access.

2. Access Card Department Processes

2.1. Application Process

2.2. Verification Process - The procedure used by the department to verify a student's/staff identity and eligibility and make sure they fulfil the requirements to be issued an access card.

2.3. Card Production Process – Once the student or staff member's information is verified, the next step is to physically create access cards. This process includes encoding user information onto the card, printing identification details, and incorporating security features to prevent tampering or duplication.

2.4. Card Distribution Process – This is the part where the access cards are checked by the member of the card production department if everything is correct, then the cards are given to the students.

2.5. Record-Keeping Process - The department uses a methodical process to keep accurate records of applications, approvals, issuances, deactivations, and replacements for access cards. This ensures compliance with data protection regulations and makes it easier to fulfil project requirements for auditing and reporting.

4.2 Investigate the technology and infrastructure

- **Technology**

- Card5 Software (Ltd, n.d.)

Market-Leading Design Software: With its extensive feature set and user-friendly interface, Card5, widely known as the most potent card design software available worldwide, sets the standard for the industry. All users, regardless of their skill level, can easily acquire and operate the system thanks to its simple setup and user-friendly design.

Comprehensive Card Design Control: Users have total command over their printing environment, enabling them to alter every element of their card designs to satisfy particular specifications. One of Card 5's important components is versatility, which allows users to add different components for distinctive card effects, like overlay, holograms, and scratch-off panel ribbons.

Database Integration: Card 5 easily connects card designs to databases such as Access, Oracle, SQL, and others, this enables effective integration of data and management. Users can encode and print cards in one go thanks to the encoding features that support both smart and magnetic stripe cards.

Integration and Compatibility: Users can import data like names, photos, and departments straight into their card designs thanks to easy database integration made possible by ODBC drivers. Strong imaging features enable real-time imaging and integration with digital cameras, giving users more design options.

Security Features: For users' peace of mind and data protection, enhanced security features like the Keyfive dongle and login authentication are available. For larger

organizations or multiple sites, network licenses are available, offering scalability and flexibility to meet changing business needs.

- **ITS**

As the main system until the switch to SAP is finished, DUT currently manages institutional information using ITS, they have been using it from the beginning and it was and still is working but there are now better options that do the job better and more efficiently.

- **Infrastructure**

- **Webcam:**

Utilized webcam specifications include:

1080p resolution for high-definition image capture.

24-bit true color at 25 FPS (frames per second) for smooth video output.

Built-in microphone for audio recording.

Manual focus and four front-facing LED lights for precise and well-lit imaging.

Plug-and-play functionality for easy setup and compatibility with various devices.

Mountable on tripods, laptops, monitors, and TVs for versatile positioning.

No on-board storage, ensuring data privacy and security.

Viewing angle: Diagonal - 90 degrees, Horizontal - 80 degrees.

- **Computers:** Lenovo or Dell towers with 8GB RAM are used by DUT as workstations for a variety of administrative and card production tasks.

Building: The card production department operates within a single building, which serves as the central hub for all access card-related activities. The building is crowded, small, and doesn't have enough lighting to produce the best images.

- **HID FARGO HDP5000 Printer:** (HID Global Corporation, 2024)

Retransfer Printing Technology: This method produces consistently excellent, high-definition printing results by using retransfer printing technology. provides exceptional 300 dpi printing resolution, guaranteeing clear text and colorful images for ID cards.

Versatile architecture: Allows for instant expansion or reduction of production capacity, giving flexibility to meet the present and future card application needs. Supports additional modules for lamination, encoding, and dual-sided printing to meet different production, durability, and security needs.

Enhanced System Capacity: Provides an optional dual-input card hopper for effectively handling larger card volumes and a variety of card types. permits one-step, inline printing and encoding with a variety of field-upgradable encoder options through seamless integration.

High-quality output: satisfies the exacting specifications of ID card applications by providing visually stunning, high-quality credentials on demand. guarantees outstanding color accuracy and print clarity, boosting ID cards' overall polished appearance.

Built-in Security Features: Provides built-in password protection, AES-256 encryption, and resin scramble data protection to safeguard sensitive cardholder information.

Optional locks further enhance security, offering organizations peace of mind and compliance with data protection regulations.

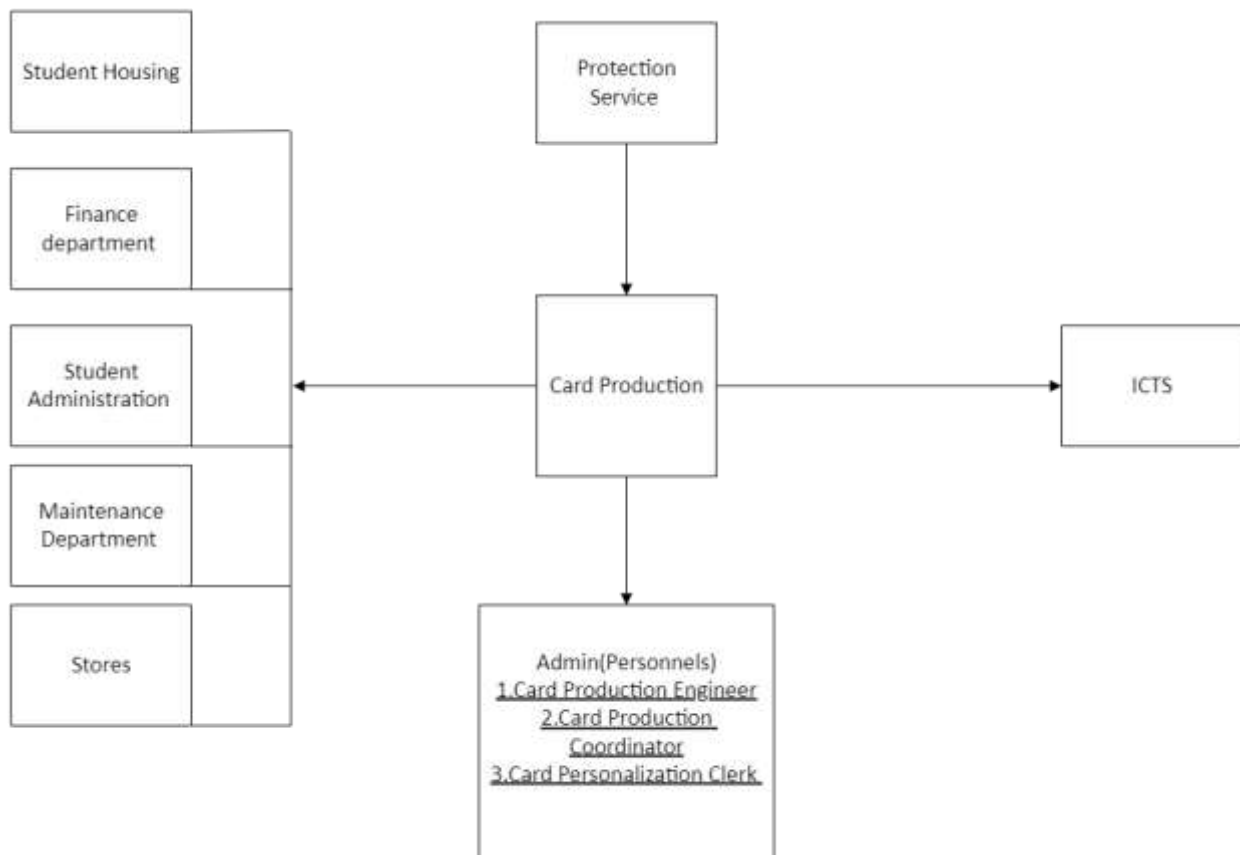
4.3 Interrogate the policies that influence the operations of the business:

Access Control Policy: This policy specifies who is allowed to access what areas or resources in the organization and how they can be granted or revoked authority to do so.

Information Security Policy: It governs the protection of sensitive information, including access control mechanisms to inhibit unauthorized disclosure or modification of stored data in access cards.

Privacy Policy: This policy provides students with whom we collect some of their personal information as we issue the card; As concerning the law, it shows rules and safety measures put so that the information is safe and prevents unauthorized collection.

4.4 Provide the business architecture.



The diagram outlines a structured workflow within the Durban University of Technology (DUT), specifically within its Protection Services department. This department holds the responsibility of ensuring the safety and security of the university's community members and its assets.

Within the Protection Services department, there exists a subunit known as the Card Production department. This segment focuses on the creation and distribution of access cards, crucial for regulating entry and ensuring security across various university premises.

A significant aspect of the Card Production department's operation is its data-sharing agreement with the Information and Communication Technology Services (ICTS) department. ICTS serves as a centralized hub for storing extensive data related to both students and staff members. This repository includes pertinent information required for producing personalized access cards, such as identification details and access privileges.

The symbiotic relationship between the Card Production and ICTS departments ensures seamless access to essential data, facilitating the efficient generation of access cards for students and staff members alike.

Furthermore, the Administration department plays a vital role in maintaining accurate records of registered students within the university system. This accurate recordkeeping is pivotal for ensuring that all individuals entitled to access cards receive them promptly and with correct information. The data managed by the Administration department serves as a foundational source for other departments, ensuring consistency and reliability across the university's operational functions.

Moreover, the Stores department collaborates with the Card Production department by providing necessary raw materials essential for the production of student access cards. This streamlined supply chain ensures that the Card Production department has access to all required resources, minimizing delays and optimizing production efficiency.

The admin Personnel includes. Card Production Engineers may be involved in the design, optimization, and maintenance of card production systems and equipment, as well as process improvement initiatives. Card Production Coordinator: Coordinators are responsible for managing the logistics and coordination of various tasks within the card production workflow, ensuring that production timelines are met and resources are utilized effectively. Card Personalization Clerk: Individuals in this role are typically responsible for personalizing cards with individual user information, such as names, ID numbers, and access privileges, using specialized printing or encoding techniques.

4.5 List the assets

Physical Assets: Devices for access control (keypads, biometric scanners, and card readers) Webcams for taking pictures of applicants; server gear for databases and access control software. Hardware for networking (cables, switches, and routers). Key fobs and access cards. Smart cards or hardware tokens for authentication

Digital Assets: Software for access control. Software for processing cards comparable to ITS Card 5 database holding system configurations, access logs, and cardholder data. Subscriptions and software licensing. Handbooks and documentation

for both software and hardware systems. Interfaces that facilitate integration across identity management, security, and HR systems

Financial Assets: Allocate funds for the acquisition of hardware, software, and licensing financing for employee salaries and training initiatives improvements to access control systems, and investments in security measures Funds set aside for unforeseen costs or crises.

5. Possible Solutions

5.1 Rejected Solutions

5.1.1 Building Access Card Production Departments in All Campuses:

This solution will require DUT to replicate its infrastructure, procedures and staff in all its campuses around Durban. Now each campus will have its own facility for making and issuing access cards to staff and students.

Advantages:

Convenience: Students and staff will no longer have to travel to get their access cards, they will be able to get them from the comfort of their campuses, which in turn improves the overall satisfaction of both the students and staff.

Reduced Workload: Having only one central location for access card issuance causes employee burnout, distributing the workload across all campuses will relieve the pressure from the employee thus creating a healthy working environment

Disadvantages:

High Initial Investment: Building Access Card Production Departments in all campuses requires a very large initial capital investment for staff, equipment and infrastructure.

Operational Complexity: When the card production department is on all campuses, there will be a lot of operational complexities which will include standardization and coordination.

Reason for Rejection:

Because of the high initial cost, operational complexity, and continuous resource requirements, it is not practical to build access card production departments on every campus.

5.1.2 Outsourcing Card Production:

DUT will have to hire a third-party service provider to produce access cards on the university's behalf. This supplier would be in charge of creating, producing, and giving staff and students access cards, so basically the university wouldn't have to do anything.

Advantages

Cost Savings: With this solution, the university won't have to invest in infrastructure, Software or staff since everything will be done by the third-party

Expertise: With all the experience a third party has, they will offer expertise and efficiency in all the processes of access card making and issuing

Scalability: With this solution adjusting to the demand is not a problem, the university won't have to worry about upgrading anything.

Disadvantages

Loss of Control: When outsourcing card production, all control is given to the third party, the quality, processes and standards are now dependent on the third party

Confidentiality Concerns: With students and staff data being shared with a third party, that causes concerns about data privacy and creates risks of the data being stolen and sold on the black market

Reason for Rejection:

There are dependency risks associated with this solution, a loss of control, and confidentiality issues if card production is outsourced. Controlling the production process is crucial because access cards are crucial for identity management and campus security. Furthermore, rather than depending on outside suppliers for essential services, the university might find it more advantageous to invest in developing internal capabilities.

5.2 Recommended Solutions

5.2.1 Upgrade Physical Infrastructure

To increase productivity and user experience, the card production department should upgrade its physical infrastructure.

Feasibility Considerations:

Business Feasibility: increases user satisfaction and efficiency. This will need a significant amount of capital.

Technical Feasibility: Renovation and installation may cause disruption and chaos within the institution.

Financial Feasibility: first expenditure for infrastructure improvements. Possible cost reductions in the long run due to increased effectiveness.

Why recommend it?

There are several advantages to modernizing the card production department's physical infrastructure, including increased cost-effectiveness, scalability, user experience, and efficiency. Since it is in line with the department's aims and objectives, it is advised as the course of action for resolving the issues raised.

5.2.2 Online Card Acquisition System

The Online Card Acquisition System redefines the card acquisition process by introducing an automated online platform for staff and students to register and create their access cards remotely. This system offers ease and efficiency, permitting users to complete the registration process from any location with internet access. Upon successful registration, users are provided with a collection date and time for their access cards, guaranteeing a seamless and organized pickup process. The online platform is made to be user-friendly, with intuitive navigation and language options available to accommodate diverse user preferences. This enhances accessibility and usability, making the registration process straightforward and easy.

Feasibility Considerations:

Business Feasibility: Putting in place an online card acquisition system makes things more convenient and helps to cut down on bottlenecks in the card issue process. The

advantages in terms of user satisfaction and efficiency exceed the initial difficulties, even though modifications to current procedures might be necessary.

Technical Feasibility: The creation of a safe web interface and its integration with the current system are essential to the project's success. It is possible to effectively handle the technical aspects and guarantee a flawless user experience with careful planning and execution.

Financial Feasibility: Even though the early phases of development incur expenses up front, the long-term advantages of decreased manual labour and enhanced productivity result in significant cost savings. Furthermore, the online system's scalability permits future expansion and fluctuations in demand without necessitating large infrastructure expenditures.

Why We Recommend It?

The implementation of an online card acquisition system offers numerous advantages that make it the most viable solution for addressing the challenges faced by the card production department.

By automating steps, the online system minimizes the need for human data entry and speeds up the card issuance process. Process simplification and productivity gains result in quicker turnaround times and better service delivery.

Future growth and demand variations can be accommodated by the online system. Its scalability allows for efficient management of growing volumes of card issuance requests without the need for large infrastructure investments or additional resources, so it makes more sense that the university takes this solution since the university has plans for future growth.

5.2.3 Card Refresh

The card refresh solution requires students to create their access cards once, only in their first year of study. After that their access cards are automatically reactivated each year, saving them from having to create new ones every year.

The card production department will use an extensive database system that keeps track of every student's enrolment status and card activation history to streamline this process. Returning students' cards are immediately reactivated upon registration,

making use of pre-existing data to streamline procedures and shorten processing times.

Students who are not in their first year only need to create new access cards when they lose their existing ones. By focusing card production efforts on these specific groups, the department optimizes efficiency and minimizes queues during peak registration periods.

Feasibility Considerations:

Business Feasibility: By reducing duplicate card creation and giving first-year students and card replacement cases priority over other tasks, the "Card Refresh" solution improves operational efficiency. It supports the department's objective of streamlining procedures while delivering high-quality service.

Technical Feasibility: Robust database management systems that can track student enrolment status and card activation history are necessary for the solution's implementation. Accurate data management and smooth operation depend on integration with current systems.

6. Analysis of Costs and Benefits

6.1 upgrade physical infrastructure

BENEFITS:

	Year 1	Year 2	Year 3	Total
Benefits (Rands)				
Scalability	25,000	20,000	15,000	60,000
Enhanced user experience	15,000	12,000	10,000	37,000
Improved service delivery	20,000	18,000	15,000	53,000
Increased Productivity	30,000	25,000	20,000	75,000
Potential Cost Reductions	18,000	20,000	22,000	60,000
Total Benefits	108,000	95,000	83,000	285,000

COSTS:

	Year 1	Year 2	Year 3	Total
Costs (Rands)				
Capital Investment	120,000	100,000	80,000	300,000
Hardware and Software purchase costs	80,000	60,000	40,000	180,000
Installation Costs	40,000	30,000	20,000	90,000
Staff Costs	50,000	40,000	30,000	120,000
Temporary Disruption Costs	15,000	10,000	5,000	30,000
Maintenance and support costs	20,000	15,000	10,000	45,000
Total Costs	325,000	255,000	185,000	765,000

Investment Appraisal:

NPV Calculation: $\text{Cash flow} / (1 + i)^t - \text{initial investment}$

Year 1: $108,000(1+0.1)^{-1} = \underline{R98,181.82}$

Year 2: $95,000(1+0.1)^{-2} = \underline{R76,446.28}$

Year 3: $82,000(1+0.1)^{-3} = \underline{R60,082.64}$

Total NPV = $R98,181.82 + R76,446.28 + R60,082.64 = \underline{R234,710.74}$

6.2 Online Card Acquisition system

BENEFITS:

	Year 1	Year 2	Year 3	Total
Benefits (Rands)				
Scalability	30,000	25,000	20,000	75,000
Increased efficiency	35,000	30,000	25,000	90,000
Enhanced business processes through Innovation	25,000	20,000	15,000	60,000
Improved support for core business functions	20,000	18,000	15,000	53,000
Increased revenue	25,000	30,000	35,000	90,000
Total Benefits	135,000	123,000	110,000	368,000

COSTS:

	Year 1	Year 2	Year 3	Total
Costs (Rands)				
Initial System Development costs	150,000	120,000	100,000	370,000
Software and Hardware purchase costs	100,000	80,000	60,000	240,000
Installations Costs	50,000	40,000	30,000	120,000
Support and Maintenance costs	25,000	20,000	15,000	60,000
Migration Costs	15,000	10,000	5,000	30,000
Training costs	30,000	25,000	20,000	75,000
Total costs	370,000	295,000	230,000	895,000

Investment Appraisal:

NPV Calculation: $\text{Cash flow} / (1 + i)^t - \text{initial investment}$

Year 1: $135,000(1+0.1)^{-1} = \underline{\underline{R122,727.27}}$

Year 2: $123,000(1+0.1)^{-2} = \underline{\underline{R90,000.00}}$

Year 3: $110,000(1+0.1)^{-3} = \underline{\underline{R72,727.27}}$

$$\text{Total NPV} = R122,727.27 + R90,000.00 + R72,727.27 = \underline{R285,454.54}$$

6.3 Card Refresh

BENEFITS:

	Year 1	Year 2	Year 3	Total
Benefits (Rands)				
Cost Saving	25,000	20,000	15,000	60,000
Enhanced operational efficiency	30,000	25,000	20,000	75,000
Reduced employee burnout	20,000	18,000	15,000	53,000
Resource Optimization	15,000	12,000	10,000	37,000
Total Benefits	90,000	75,000	60,000	225,000

COSTS:

	Year 1	Year 2	Year 3	Total
Costs (Rands)				
Initial development Costs	90,000	70,000	50,000	210,000
Maintenance Costs	30,000	25,000	20,000	75,000
Process Modification Costs	20,000	15,000	10,000	45,000
Total Costs	140,000	110,000	80,000	330,000

Investment Appraisal:

NPV Calculation: $\text{Cash flow} / (1 + i)^t - \text{initial investment}$

$$\text{Year 1: } 90,000(1+0.1)^{-1} = \underline{R81,818.18}$$

$$\text{Year 2: } 75,000(1+0.1)^{-2} = \underline{R61,983.47}$$

$$\text{Year 3: } 60,000(1+0.1)^{-3} = \underline{R49,587.62}$$

$$\text{Total NPV} = R81,818.18 + R61,983.47 + R49,587.62 = \underline{R193,389.27}$$

7. Conduct an impact assessment for each of the options

- Card refresh

Has a Positive impact on the Card Production Unit , since it allows for use of existing students and staff registering for the new year to carry over their details onto the next year . It impacts the financial aspects of the company in a positive way , not only will cost of salaries drop , but with less stress on the staff this will result in higher quality work and less human error due to pressure and deadlines .

It will also have a positive impact the speed of the business process of the Production Unit . Verifying and refreshing existing students and staff members on the database allows for a more happier consumer base reaction thus impacting the overall image of the Production Unit in a positive way. .

Another positive impact that occurred in this solution is that with the reduced redundancy of making a new card, physical materials such as cards, ink, and lanyards won't have to be bought as often resulting in the initial cost at the start of the year to be much less on assets. The reduction of non-first-year card productions will also positively impact the water and electricity paid by the department resulting

in a decrease of the total cost every year by almost 66%.

○ Online Card Acquisition system

Has a positive and a negative impact on the Card Production unit.

Positive Impacts :

Pushing students and staff to register their cards online before physically collecting them allows for the Card Production Unit to experience less stress on registering user cards themselves thus increasing the energy levels of the employees as well as saving them time. This positively impacts the overall speed and efficiency of the system.

Faster processes then results to more satisfied students and staff, which greatly enhances the reputation of DUT services.

An Online presence greatly reduces human error while increasing consistency in producing student and staff member's cards.

Negative Impacts :

Some students enrolled into DUT come from a less techno-savvy background who might experience difficulties in registering online .Users who might not have access to internet or never owned a mobile device might miss the chance to register their cards online .

Another Major Negative impact is the costs needed to operate a website online. The Card Production Unit will have to pay constantly huge amounts of money to run the site. IT will also cost the unit to hire an IT professional to help with crashes, updates, maintenance and someone to manage the database of all created cards to avoid reputation. This will result in an increase in Wages.

This Unit will also have to invest in Training for the current staff to be familiar with the website and help create awareness of the site itself, to avoid learners mistakenly coming to register physically.

○ **Upgrade physical infrastructure**

Upgrading the physical infrastructure has a Negative impact on The Financial aspect of the Card Production Unit. It will cost significant amounts of resources and time to renovate the area for expansion and to purchase more equipment such as monitors and printers.

It will also have a negative impact on performance due to new unfamiliar Hardware brought into the system .

The system will also need legal permits in order to renovate and remove environmental factors such as trees , bushes and animal habitats , which will cost the Unit to obtain .

8. Conduct a risk assessment for each of the options

- Card refresh

Technical Risk: The difficulties in maintaining a database may result in inaccurate data or malfunctions in the system, which could cause problems with card activation procedures. Software compatibility problems or insufficient data migration techniques may make this risk worse.

Countermeasure: Create and put into place reliable data validation procedures, such as automated checks and manual verification procedures, to maintain the integrity and validity of the data kept in the database. Before deploying the software fully, make sure all compatibility issues are found and fixed through comprehensive testing and pilot runs. To reduce the possibility of errors during the switch to the new system, create thorough data migration plans that include data cleansing and validation processes.

Financial Risk: The card refresh system's initial setup costs and continuing database management expenses will definitely go over budget. Unexpected software licensing costs or hardware upgrades could put a strain on finances.

Countermeasure: To accurately determine the card refresh system's total cost of ownership, including setup and ongoing operating costs, perform a thorough cost analysis. Examine ways to cut costs, such as by using open-source software to minimize licensing costs or negotiating bulk discounts with suppliers. To keep an eye on spending and spot possible cost overruns early on, establish strict budgetary controls and conduct regular financial reviews. To handle unforeseen costs and guarantee financial stability throughout the project lifecycle, set aside money in reserve accounts or contingency funds.

Operational Risk: Relying too much on automated card activation procedures increases the possibility of system malfunctions, which could prevent users from

accessing facilities and services and cause operational inefficiencies and user unhappiness.

Countermeasure: Create strict risk mitigation plans. Establish reliable systems for monitoring and alerting to detect and resolve performance problems before they become major disruptions to the department. Regularly test emergency protocols and make sure staff members are prepared to handle unforeseen circumstances by conducting frequent training and simulating what could happen and how best to prevent it. Work together to create coordinated response plans and create open channels of communication during emergencies with important stakeholders.

-Online Card Acquisition system

Scalability challenges- scalability is one main concerns in systems, if the system cannot handle large volumes of users during peak periods this could result in slow performance and system crashes.

System Downtimes- technical issues or maintenance may cause system downtime which prevents users from accessing the system and causing frustration.

Security risks - online systems are vulnerable to cybersecurity threats, these includes Malware infections, DoS attacks ,data breaches, which can compromise sensitive information, damage reputation and cause disruption in operation.

Counter Measures: the system must be designed horizontally or vertically to manage increasing user traffic during peak times

Counter measures: to insure high availability and fault tolerance at peak times, implementation of failover schemes and redundant components, this includes redundant servers, data replication, and automated failover schemes.

- Upgrade physical infrastructure

Upgrades of physical infrastructure may cause compatibility issues if not researched properly as new infrastructure may not be compatible with existing systems and software which might results in potential system failures and integration challenges. Furthermore, there could be disruption in productivity as a results of infrastructure upgrades affecting the normal business operations. Operational risks might also occur because of new equipment being introduced to staff members causing adaptability challenges.

Counter measures: testing and piloting of the new infrastructure before deployment will help to identify and address any compatibility issues and technical challenges. Development of a detailed plan for the infrastructure upgrades, including timelines, budget constraints, and contingency plans will help to minimize operational disruptions. Investment in staff training and skills development can help to mitigate the risks of adaptability.

9. Conclusion/Recommendations

9.1 Summarise the business case and set out the decisions that senior managers are asked to take.

The DUT community has grown over the years, as a result the institution has made protecting its assets and students/staff a top priority. The institution has a sub-unit under the Protection Service Unit (PSU) called Card Production because of how important safety is. The primary duty of card production is to create access cards that provide simple access control to the institution for personnel, students, and pertinent subcontractors. We made the decision to look into the business's procedures in order to identify the underlying causes of the long line issue and provide recommendations and solutions to the department in response to the lengthy line issue the institution was experiencing during the card production period. The card production department must advance in its capacity to deliver staff and student access cards in a timely and reliable way, given that the demand for these cards surges at the start of each academic year. This will stop pupils from having to wait in long lines to receive access cards at the start of each academic year. After doing research and consulting with relevant parties, it was determined that improving the physical infrastructure, implementing an online card acquisition system, and replacing cards might potentially address the current issue. On the other hand, it is advised that the department use option 2—an online acquisition system. This system is user-friendly and effective; it allows staff and students to register and create their access cards remotely through an automated web platform, which can be accessed from any location with internet access.

9.2 Gantt Chat

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10. Appendices and supporting information

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